Fair Best Arm Identification with Fixed Confidence

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1 Introduction

In sequential decision-making, the Multi-Armed Bandit (MAB) problem is a foundational framework in which a learner repeatedly selects among K arms, observing stochastic rewards, with the goal of maximizing some objective over time. A particularly important variant is Best Arm Identification (BAI) [3], where the objective is not to maximize cumulative reward but rather to identify the arm with the highest expected reward as quickly and reliably as possible.

While BAI has been extensively studied, classical algorithms do not account for fairness — the requirement that each arm receives a minimum share of selections during the identification process. This is a meaningful limitation in practice. Consider a hospital testing K treatment protocols: a pure BAI algorithm will concentrate samples on the most promising treatments, potentially leaving some protocols nearly untested. In wireless scheduling, a base station identifying the best user equipment (UE) to serve must still guarantee minimum throughput to all UEs. In both cases, ignoring fairness can lead to outcomes that are statistically efficient but ethically or legally problematic.

This project studies *Fair Best Arm Identification* (F-BAI), a framework introduced by Russo and Vannella [1] that incorporates fairness constraints directly into the BAI problem. The framework imposes a lower bound on the selection rate of each arm, either as a fixed pre-specified value or as a function of the unknown model parameter. The central contribution of the paper is F-TaS (Fair Track-and-Stop), an algorithm that provably achieves optimal sample complexity while satisfying these fairness constraints.

Our project has two objectives. First, we implement F-TaS and reproduce the synthetic experiments from Section VI-A of [1], evaluating sample complexity and fairness violation across agonistic and antagonistic fairness settings under both pre-specified and ε-dependent constraints. Second, we propose and empirically examine a modification to the algorithm, investigating how quickly fairness violations decay as the confidence parameter $\delta \to 0$ for ε-dependent constraints. Both contributions are empirical in nature; we do not prove new theoretical results but rather validate and extend the experimental findings of [1].

2 Background

2.1 Best Arm Identification

The problem of Best Arm Identification under fixed confidence was formalized by Audibert et al. [3] and later studied in depth by Kaufmann et al. [4]. The goal is to identify the best arm a^* with probability at least $1-\delta$, using as few samples as possible. The key insight from [2] is that the sample complexity of any δ-PAC algorithm is lower bounded by a problem-specific constant T^*, scaled by $\log(1/\delta)$. Their Track-and-Stop (TaS) algorithm asymptotically achieves this lower bound, making it the state-of-the-art for BAI with fixed confidence.

The core technique in TaS is to track the optimal sampling allocation w^* — the distribution over arms that minimizes the characteristic time T^* — using a certainty-equivalence principle: at each round, compute w^* using the current empirical estimates $\hat{\mu}(t)$ in place of the unknown true means μ, then sample accordingly.

2.2 Fairness in Bandits

Fairness has been studied extensively in the regret minimization setting [5, 6, 7]. The main notions include pre-specified fairness, which constrains arm selection rates to lie within a fixed range, and individual fairness,

which requires that similar arms be treated similarly. Most existing works focus on regret minimization, where the goal is to maximize cumulative reward. The BAI setting, where the goal is pure exploration, has received much less attention from a fairness perspective.

To the best of our knowledge, [1] is the first work to systematically study fairness in BAI with fixed confidence. The only prior work in this direction is [8], which considers fairness constraints on subpopulations rather than on individual arms, making it a fundamentally different setting.

2.3 Contribution of This Paper

The key contributions of [1] relative to prior work are:

1. A formal framework for F-BAI that encompasses multiple classical notions of fairness, including proportional fairness and individual fairness.
2. An instance-specific sample complexity lower bound for any p-fair δ-PAC algorithm, and a quantification of the *price of fairness* — the additional samples required due to fairness constraints.
3. The F-TaS algorithm, which asymptotically matches the lower bound while satisfying fairness constraints at every round (for pre-specified constraints) or asymptotically (for ε-dependent constraints).

3 Problem Setup

3.1 Bandit Model

We consider a stochastic bandit problem with K arms, denoted $[K] = \{1, \dots, K\}$. At each round $t \geq 1$, the learner selects an arm $a_t \in [K]$ and observes a reward $r_t \sim \mathcal{N}(\mu_{a_t}, 1)$. Rewards are independent across rounds, and $\mu = (\mu_a)_{a \in [K]}$ is the unknown parameter vector. We denote by $a^* = \arg\max_a \mu_a$ the unique best arm, and define the sub-optimality gap for arm a as $\Delta_a = \mu_{a^*} - \mu_a$.

A BAI algorithm consists of three components: a *sampling rule* π_t that selects which arm to pull at round t, a *stopping rule* τ_δ that decides when to stop, and a *decision rule* that outputs a guess $\hat{a}$ of the best arm.

3.2 Fairness Constraints

In Fair BAI, the learner must identify a^* while satisfying a set of fairness constraints on arm selection rates. Let $N_a(t) = \sum_{s=1}^{t} \mathbf{1}\{a_s = a\}$ denote the number of times arm a has been selected up to round t, and let $w_a = \mathbb{E}[N_a(\tau_\delta)]/\mathbb{E}[\tau_\delta]$ denote the asymptotic selection rate. The fairness constraints take one of two forms:

Definition 1 (Pre-specified constraints)**.** *The selection rate at stopping must satisfy*

$$\frac{\mathbb{E}[N_a(\tau_\delta)]}{\mathbb{E}[\tau_\delta]} \geq p_a, \quad \forall a \in [K], \tag{1}$$

where $p = (p_a)_{a \in [K]}$ is a fixed, model-agnostic fairness vector with $\sum_a p_a \leq 1$.

Definition 2 (ε-dependent constraints)**.** *Asymptotically as $\delta \to 0$, the selection rate must satisfy*

$$\liminf_{\delta \to 0} \frac{\mathbb{E}[N_a(\tau_\delta)]}{\mathbb{E}[\tau_\delta]} \geq p_a(\mu), \quad \forall a \in [K], \tag{2}$$

where $p_a(\mu) : \mathbb{R}^K \to [0, 1]$ is a continuous function of the unknown model parameter.

An important special case of ε-dependent constraints is proportional fairness, where $p_a(\mu) = p_0 \cdot \mu_a / \sum_b \mu_b$ for some $p_0 \in [0, 1]$. This requires arms with higher mean rewards to be sampled more frequently, proportional to their merit.

Definition 3 (p-fair δ-PAC algorithm)**.** *An algorithm is p-fair δ-PAC if it satisfies the fairness constraint in (1), outputs the correct best arm with probability at least $1 - \delta$, and stops almost surely.*

3.3 Sample Complexity Lower Bound

The fundamental difficulty of F-BAI is captured by the *characteristic time*:

$$T_p^* = 2 \inf_{w \in \triangle_p} \max_{a \neq a^*} \frac{w_a^{-1} + w_{a^*}^{-1}}{\Delta_a^2}, \tag{3}$$

where $\triangle_p = \{w \geq p : \sum_a w_a = 1\}$ is the clipped simplex enforcing the fairness constraints.

Theorem 1 (Lower bound, [1])**.** *Any p-fair δ-PAC algorithm satisfies*

$$\mathbb{E}[\tau_\delta] \geq T_p^* \cdot \log(1/2.4\delta). \tag{4}$$

The intuition behind T_p^* is as follows. To rule out arm a as the best arm, you need enough statistical evidence from both arm a and arm a^*. The term $w_a^{-1} + w_{a^*}^{-1}$ captures how much information you have about this pair, and Δ_a^2 captures how easy they are to distinguish. The infimum over $w \in \triangle_p$ finds the best allocation subject to fairness — but unlike unconstrained BAI, the optimizer is restricted to $\triangle_p$, which can only increase T_p^* relative to the unconstrained case.

3.4 Price of Fairness

The ratio T_p^*/T^* quantifies how much additional sampling fairness requires. The paper establishes:

Lemma 1 (Price of fairness, [1])**.**

$$1 \leq \frac{T_p^*}{T^*} \leq O\left(\min\left\{\frac{1}{1 - p_{\text{sum}}},\ \frac{1}{K p_{\min}}\right\}\right), \tag{5}$$

where $p_{\text{sum}} = \sum_a p_a$ *and* $p_{\min} = \min_{a: p_a > 0} p_a$.

There are two regimes. When p_{sum} is close to 1, fairness consumes most of the sampling budget and leaves little room for optimization, driving up the price. When $p_{\min}$ is very small but K is large, forcing exploration of arms that would naturally receive near-zero allocation becomes costly. The price is governed by whichever term is the binding constraint.

4 The F-TaS Algorithm

F-TaS (Fair Track-and-Stop) is designed to asymptotically achieve the lower bound $T_p^* \cdot \log(1/\delta)$ while satisfying the fairness constraints. It extends the Track-and-Stop algorithm of [2] to the fair setting.

4.1 Sampling Rule

The core idea is the *certainty-equivalence principle*: since the optimal allocation w_p^* depends on the unknown μ, F-TaS substitutes the current empirical estimates $\hat{\mu}(t)$ and tracks the resulting estimate $\hat{w}_p^*(t)$.

At each round t, F-TaS computes:

$$\hat{w}_p^*(t) = \arg \inf_{w \in \triangle_p} \max_{a \neq \hat{a}^*(t)} \frac{w_a^{-1} + w_{\hat{a}^*(t)}^{-1}}{\hat{\Delta}_a(t)^2}, \tag{6}$$

where $\hat{a}^*(t) = \arg\max_a \hat{\mu}_a(t)$ and $\hat{\Delta}_a(t) = \hat{\mu}_{\hat{a}^*(t)}(t) - \hat{\mu}_a(t)$.

The actual sampling distribution mixes this with a forced exploration policy:

$$\phi(t) = (1 - \varphi_t)\, \hat{w}_p^*(t) + \varphi_t\, \phi_c, \quad \varphi_t = \frac{1}{2\sqrt{t}}. \tag{7}$$

The forced exploration policy ϕ_c ensures every arm is sampled infinitely often, keeping the estimates $\hat{\mu}(t)$ reliable throughout. For pre-specified constraints, $\phi_{c,a} = p_a$ for arms with $p_a > 0$, since the fairness floor already guarantees sufficient exploration. For ε-dependent constraints, $\phi_{c,a} = 1/K$ (uniform), since no pre-specified floor exists.

The mixing weight $\varphi_t = 1/(2\sqrt{t})$ shrinks to zero over time, so the forced exploration term becomes negligible asymptotically and does not affect the sample complexity in the limit.

4.2 Stopping Rule

F-TaS stops at the first round t where the generalized likelihood ratio statistic exceeds a threshold:

$$Z(t) = \frac{t}{2T_p^*(t)} \geq \beta(\delta, t), \tag{8}$$

where $T_p^*(t) = 2\max_{a \neq \hat{a}^*(t)} \frac{(N_a(t)/t)^{-1} + (N_{a^*}(t)/t)^{-1}}{\hat{\Delta}_a(t)^2}$ uses the empirical allocation $N_a(t)/t$ rather than the target $\hat{w}_p^*(t)$, measuring how much evidence has actually accumulated. The threshold used in the experiments is $\beta(\delta, t) = \log((\log(t) + 1)/\delta)$, following [2].

4.3 Guarantees

Theorem 2 (Optimality and fairness of F-TaS, [1])**.** *F-TaS is p-fair δ-PAC (resp. asymptotically $p(\mu)$-fair δ-PAC for ε-dependent constraints), and satisfies:*

$$\limsup_{\delta \to 0} \frac{\mathbb{E}[\tau_\delta]}{\log(1/\delta)} \leq T_p^*. \tag{9}$$

Combined with the lower bound, this means F-TaS is asymptotically optimal: no algorithm can do better in the $\delta \to 0$ regime.

5 Experiments

5.1 Setup

We reproduce the synthetic experiments from Section VI-A of [1]. We compare three algorithms:

- **F-TaS**: the proposed fair algorithm
- **TaS**: the original Track-and-Stop [2], with no fairness constraints
- **Uniform Fair**: selects arm a with probability proportional to $p_a(\hat{\mu}(t)) + (1 - p_{\text{sum}})/K$, guaranteeing fairness but ignoring optimality

We evaluate across four conditions formed by two axes:

Agonistic vs. antagonistic: The fairness vector p is called agonistic when it is aligned with the unconstrained optimal allocation w^* — fairness promotes the exploration the algorithm would already want to do. It is antagonistic when p is inversely related to w^* — fairness forces over-sampling of sub-optimal arms.

Pre-specified vs. ε-dependent: Pre-specified rates are fixed constants independent of μ. ε-dependent rates are functions of μ, requiring the algorithm to learn the constraints simultaneously with the means.

For pre-specified constraints, the fairness vector is set as $p_a = p_0[\omega w_a^* + (1 - \omega)\bar{w}_a^*]$, where $\bar{w}_a^* = (1/w_a^*)/\sum_b (1/w_b^*)$ is the inverse allocation. We set $\omega = 0.9$ for agonistic and $\omega = 0.1$ for antagonistic. For ε-dependent constraints, agonistic rates are set inversely proportional to the gaps Δ_a, and antagonistic rates are set proportional to Δ_a.

We evaluate performance on two metrics:

- **Sample complexity** $\mathbb{E}[\tau_\delta]$: the expected number of rounds until stopping
- **Fairness violation** $\mathbb{E}[\max_a (p_a - N_a(\tau_\delta)/\tau_\delta)^+]$: the average maximum deviation below the required rate at stopping

Due to computational constraints, results are reported at $\delta = 0.1$ over 100 independent runs. We note that at this value of δ, all algorithms remain far from their asymptotic lower bounds, as the lower bounds scale as $\log(1/\delta)$ and the asymptotic regime requires much smaller δ. This is an inherent limitation of our evaluation.

5.2 Pre-specified Constraints

We consider $K = 10$ arms with expected rewards linearly ranging in $[0, K/2.5]$, with $p_0 = 0.9$.

Our results at $\delta = 0.1$ are consistent with the claims of [1]. We first examine the theoretical optimal allocations w^* and w^*_{fair} across the four conditions (Figure 1). In the agonistic settings, the fairness floor p lies below w^* for most arms, so w^*_{fair} tracks w^* closely — there is little tension between optimality and fairness. In the antagonistic settings, the fairness floor is set above w^* for weaker arms, pulling w^*_{fair} significantly away from w^* and introducing a meaningful price of fairness.

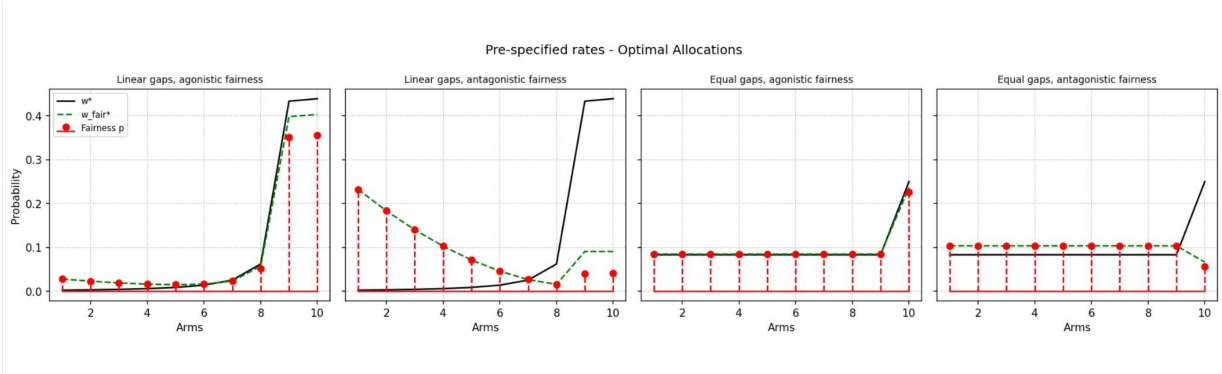


Figure 1: Theoretical optimal sampling allocations w^* (unconstrained), w^*_{fair} (fair-constrained), and fairness floor p across four experimental conditions. In agonistic settings, p lies below w^* so w^*_{fair} tracks w^* closely. In antagonistic settings, p is set above w^* for weaker arms, pulling w^*_{fair} significantly away from w^*.

Turning to sample complexity (Figure 2), the most striking result occurs in the linear gaps, antagonistic setting, where the y-axis scale reaches 4000 — substantially higher than the other three conditions. Here, TaS achieves the lowest sample complexity among the three algorithms, but does so by ignoring the fairness constraints entirely. F-TaS incurs higher sample complexity in order to remain fair, and Uniform Fair exhibits the highest variance. In the agonistic settings, F-TaS and TaS achieve comparable sample complexity, consistent with the paper's claim that the price of fairness is low when constraints are aligned with the optimal allocation. Uniform Fair is consistently higher and more variable across all conditions.

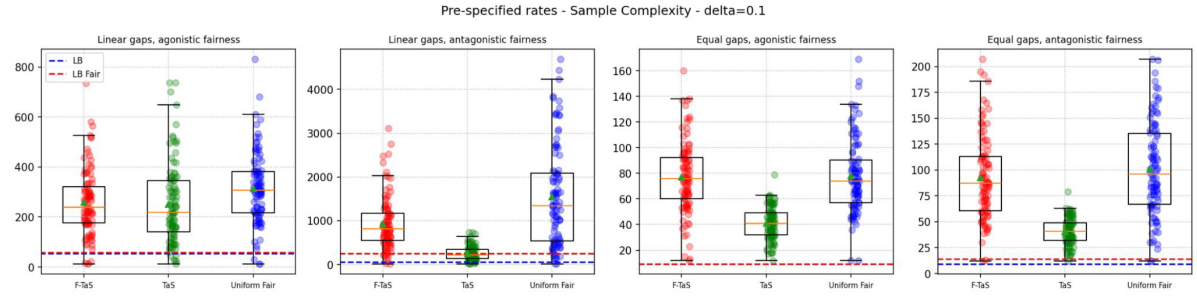


Figure 2: Sample complexity distributions at $\delta = 0.1$ across four experimental conditions. Blue dashed line: unfair lower bound $T^* \cdot \log(1/\delta)$. Red dashed line: fair lower bound $T^*_p \cdot \log(1/\delta)$. All algorithms remain far above both lower bounds at this value of δ, consistent with the asymptotic nature of the guarantees.

The fairness violation distributions (Figure 3) further clarify this tradeoff. In the agonistic setting, all three algorithms exhibit relatively low violations, though TaS has a heavier tail extending to approximately 30%. In the antagonistic setting, the separation becomes stark: F-TaS and Uniform Fair both concentrate near zero violation, while TaS exhibits violations centered around 18–20%.

Notably, Uniform Fair achieves slightly lower violations than F-TaS in the antagonistic setting. This occurs because Uniform Fair assigns equal probability $1/K$ to every arm, which happens to satisfy the

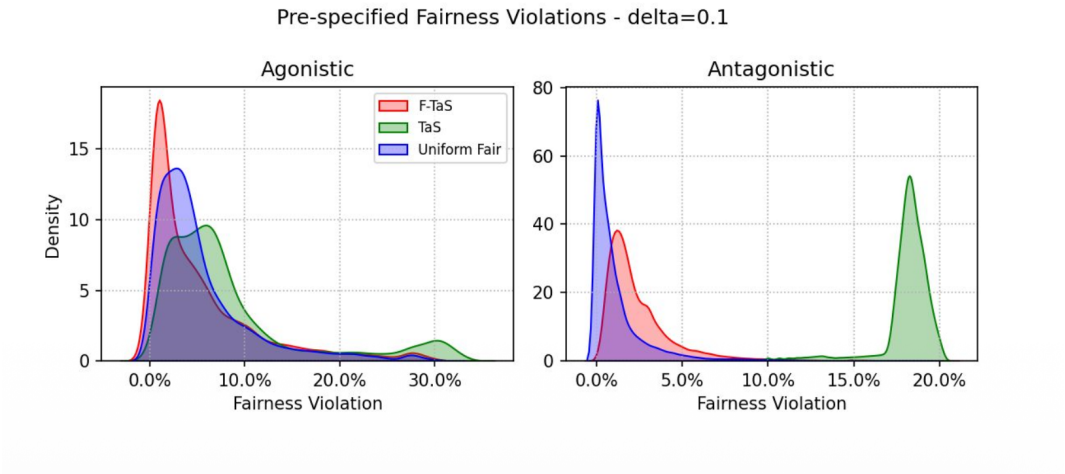


Figure 3: Fairness violation distributions at $\delta = 0.1$ for agonistic (left) and antagonistic (right) pre-specified constraint settings. F-TaS and Uniform Fair concentrate near zero in both settings. TaS exhibits violations centered around 18–20% in the antagonistic setting, confirming that it ignores the fairness constraints in favor of sample efficiency.

antagonistic fairness constraints by construction in this setting. F-TaS, being adaptive, operates closer to the fairness boundary and occasionally dips below it due to noise in its empirical estimates — a cost that does not arise in the agonistic setting where the algorithm has ample slack above the fairness floor.

5.3 ε-dependent Constraints

We consider $K = 15$ arms with rewards linearly ranging in $[0, 5]$, with $p_0 = 0.7$.

For the ε-dependent setting, our primary empirical contribution is the fairness violation convergence analysis presented in Section 6, which directly targets the asymptotic nature of the fairness guarantee. A full reproduction of the paper's Table I results for this setting was not completed due to computational constraints; we note this as a limitation.

That said, the qualitative behavior of F-TaS under ε-dependent constraints is consistent with the pre-specified case. The performance gap between F-TaS and Uniform Fair widens further in the antagonistic setting because Uniform Fair lacks the adaptive optimization that allows F-TaS to concentrate samples efficiently, and the unknown nature of the constraints makes it harder to maintain fairness while exploring optimally.

5.4 Discussion

Across all four pre-specified conditions, F-TaS is the only algorithm that achieves both competitive sample complexity and low fairness violation simultaneously. TaS is sample-efficient but frequently violates fairness, with violations exceeding 18–20% in the antagonistic setting where constraints conflict most strongly with the unconstrained optimal allocation. Uniform Fair satisfies fairness reliably but at a significant cost in sample complexity, particularly in the antagonistic setting. F-TaS navigates this tradeoff by tracking the optimal fair allocation w^*_{fair}, paying only the necessary price of fairness and no more. These results are consistent with the theoretical guarantees of [1] and confirm that the reproduction is successful at $\delta = 0.1$.

6 Modification: Fairness Violation Convergence as $\delta \to 0$

6.1 Motivation

For ε-dependent constraints, F-TaS only guarantees fairness asymptotically as $\delta \to 0$. This is because the fairness rates $p_a(\mu)$ depend on the unknown μ, so the algorithm must estimate them simultaneously with the arm means. The guarantee relies on the fact that as $\delta \to 0$, the stopping time $\tau_\delta \to \infty$, giving the empirical estimates $\hat{\mu}(t)$ time to converge to μ and thus $p_a(\hat{\mu}) \to p_a(\mu)$. For any fixed $\delta > 0$, some residual violation is expected.

While the existence of this asymptotic guarantee is established theoretically in [1], the paper does not characterize how quickly violations decay as δ decreases. Understanding this convergence rate is practically important: it tells us how small δ must be in order for the fairness guarantee to be approximately satisfied in practice.

6.2 Experimental Setup

We run F-TaS under the antagonistic θ-dependent constraint setting using $K = 15$ arms with $\theta = \text{linspace}(0, 5, 15)$ and rewards $r \sim \mathcal{N}(\theta_a, 1)$. We sweep δ over the range $\{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}\}$ and measure the mean fairness violation at each level, averaged over independent runs. To characterize the decay rate, we fit a power law to the empirical violation curve on a log-log scale.

6.3 Results

Figure 4 shows mean fairness violation as a function of δ on a log-log scale.

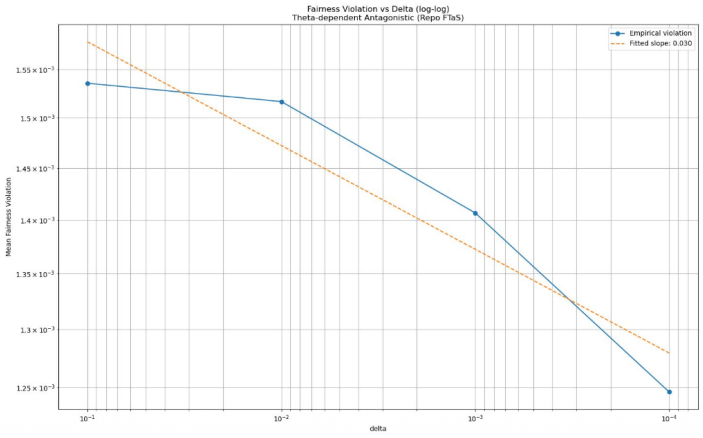


Figure 4: Mean fairness violation vs. δ on a log-log scale for the antagonistic θ-dependent setting. The empirical curve (blue) decreases monotonically, confirming the asymptotic guarantee of [1]. The fitted power law (orange dashed) has slope 0.030, suggesting violations scale approximately as $\delta^{0.030}$.

Violations decrease monotonically as δ decreases, directly confirming the paper's asymptotic guarantee. The empirical curve is well approximated by a power law with fitted exponent 0.030, suggesting that violations scale approximately as $\delta^{0.030}$ in this setting.

Two observations are worth noting. First, the absolute violations are already small throughout the range — on the order of 10^{-3}, or less than 0.2% — indicating that F-TaS satisfies fairness well even at moderate δ values. Second, the fitted slope of 0.030 is shallow, meaning violations decay slowly with δ. This suggests that while the asymptotic guarantee holds, a substantial reduction in δ is required to drive violations meaningfully lower.

6.4 Discussion

Our results are consistent with the asymptotic guarantee of [1] and provide an empirical characterization of the convergence rate that the paper does not supply. The shallow decay rate raises a practical concern: for applications where a specific fairness tolerance is required at a fixed δ, the asymptotic guarantee alone may not be sufficient to predict whether the constraint will be approximately met. We note that this experiment was conducted only in the antagonistic setting; a complete characterization would require comparing the decay rate across agonistic and antagonistic conditions, which we leave as future work.

7 Open Problems and Future Directions

The work of [1] raises several open questions that are worth highlighting.

Finite-time fairness for ε-dependent constraints. The fairness guarantee for ε-dependent constraints is asymptotic — it only holds in the limit as $\delta \to 0$. For practical applications where δ is fixed at a moderate value (e.g., $\delta = 0.1$), it is unclear how much fairness violation to expect. Our modification investigates this empirically and finds that violations decay slowly, approximately as $\delta^{0.030}$ in the antagonistic setting. A theoretical characterization of the finite-time violation rate remains an open problem.

Non-asymptotic sample complexity. The optimality of F-TaS is asymptotic in δ. Our experiments at $\delta \in \{0.1, 0.001\}$ confirm that algorithms remain far from their lower bounds at these values, with the ratio of empirical sample complexity to the fair lower bound shrinking but still substantially above 1. Deriving non-asymptotic bounds that hold for fixed δ would be valuable, particularly for understanding performance when high confidence is not required.

Extensions to structured bandits. The paper considers unstructured Gaussian bandits. Extending F-BAI to linear bandits, contextual bandits, or Lipschitz bandits — where arms share structure — is a natural and practically relevant direction. Fairness constraints in these settings could interact with the structure in non-trivial ways.

Variance estimation. The authors note that quantifying the variance of F-TaS is technically challenging. Understanding the concentration of the sample complexity around its expectation would strengthen the practical guarantees of the algorithm.

Decay rate across fairness settings. Our empirical investigation of the violation convergence rate was limited to the antagonistic θ-dependent setting. Comparing the decay rate in agonistic vs. antagonistic conditions would clarify whether the hardness of the fairness constraints affects how quickly the asymptotic guarantee becomes practically meaningful.

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